

Lab 6: Hash Tables

CSCI 41

Objectives:

- Practice hash tables by implementing our very own
- Write a templated class with *two* template parameters

Part 1: Get the starter code

First, cd into the folder where you want to store your class files (e.g., ~/labs). Copy the starter code there with the following line:

```
gimme lab06@csci41
```

This will make a lab06 folder in whatever directory you were in.

Part 2: Implement hash functions for a bunch of types

Your hash function should work with int, char, string, and double keys, so you'll need to make hash functions for all of those types in hashMap.h. The hash function must be called Hash and return an unsigned int—see the stub Hash function I've given you for ints and follow that pattern.

Part 3: Implement the HashMap class

Then, also in hashMap.h, you can start implementing the HashMap class using open hashing. There are only three methods, and the comments should guide your implementation. The (key, value) pairs that you'll insert into the hash table are of type HashTableItem.

The hash table itself is a `std::vector<std::list<HashTableItem>>` called `table`. In the constructor you'll need to initialize this vector to have a size of `tableSize`.

Part 4: Get the tests to pass and write your own

Get the tests in `hashTableTests.cpp` working, and write 2 new tests of your own. Compile with `make hashMapTests`.

Part 5: Submit your code

When you're satisfied with your solutions, you can submit them to the autograder for grading. **Do not submit and assume you got 100%—you may be unpleasantly surprised. Always check your grade.**

From your solution directory, submit your code to the autograder using the following command on the terminal:

```
turnin lab06@csci41 hashMap.h hashMapTests.cpp
```

The autograder will grade your code within a minute or so. If it's not working, please yell at Lawton to fix it. Run `view-grades` to look at your grades on the terminal, or go to the class website to view them there. You may resubmit as much as you want before the due date—just follow this same process after you've updated your programs.

Rubric

Rubric Item	Points (14 pts total)
Lawton's tests pass	10 pts (5 for each test)
Your tests pass	4 pts (2 for each test)